

PROGRAM 1:

To Install Linux VM/WSL and Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing.

1. Install Linux VM/WSL and Configure GCC

a) Ubuntu installation

Ubuntu (WSL) is already installed in the system. After opening Ubuntu, the terminal is available as shown below.

```
rishi@RISHI-VP:~$ pwd
/home/rishi
rishi@RISHI-VP:~$ ls
oslab
rishi@RISHI-VP:~$
```

b) Terminal

Basic terminal commands were used.

```
rishi@RISHI-VP:~$ pwd
/home/rishi
rishi@RISHI-VP:~$ ls
oslab
rishi@RISHI-VP:~$ cd oslab
rishi@RISHI-VP:~/oslab$ pwd
/home/rishi/oslab
rishi@RISHI-VP:~/oslab$ ls
hello hello.c
rishi@RISHI-VP:~/oslab$
```

c) Linux filesystem

Root directory and its contents were viewed.

```
rishi@RISHI-VP:~$ cd /
rishi@RISHI-VP:/ $ ls
bin boot dev etc home init lib lib64 lost+found media mnt opt proc root run sbin snap srv sys tmp usr var
rishi@RISHI-VP:~$ cd ~
rishi@RISHI-VP:~$ pwd
/home/rishi
```

d) GCC installation

GCC is already installed in the system.

```
rishi@RISHI-VP:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
rishi@RISHI-VP:~$
```

e) Compiling and executing a C program

The C program in hello.c was compiled and executed.

```
rishi@RISHI-VP:~/oslab$ cat hello.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    printf("PID = %d\n", getpid());
    return 0;
}
rishi@RISHI-VP:~/oslab$ gcc hello.c -o hello
rishi@RISHI-VP:~/oslab$ ./hello
PID = 9206
rishi@RISHI-VP:~/oslab$
```

Result: Linux (WSL) was successfully installed and configured. GCC is working properly.
A C program was successfully compiled and executed.